

CONSTRAINT: Must use the nRF24L01 module

Main Work Elements

1. **RF link bring-up** — building the STM32F411 ground-station dongle from scratch — native USB, nRF24 driver, USB bridging, with no scaffolding provided — and completing the stubbed nRF24 SPI driver on the ESP32 controller side. Ground-station hardware is USB-bus-powered from the PC, drawing nothing from the shared system power.
2. **Comms protocol design** — defining the application-layer protocol over the raw nRF24 link — framing, addressing, and any reliability mechanism such as ACK/retry — as open design space rather than a fixed specification.
3. **Dashboard software** — live monitoring of all operational telemetry and full command and control of system operations.
4. **Anti-sway motion-profile modelling** — modelling the payload/pendulum dynamics and deriving ramp-rate or velocity-shaping parameters, sent to the controller as open-loop feedforward profiles executed on MCTR I's drive.
5. **Obstacle-avoidance path planning** — the same open-loop, planned-navigation pattern, generating motion commands around known obstacles.
6. **Testing and validation** — validating dashboard behaviour against actual crane operation and confirming the anti-sway and obstacle-avoidance profiles perform as modelled.

Scoped for effort parity with the other four 2026 work packages (MCTR I, MCTR II, EEII I, EEII II, REII).